DRUGS · ALLERGY, RHINITIS AND BIOLOGICS

Omalizumab

Omalizumab is a monoclonal antibody that binds free IgE and stops it attaching to mast cells and basophils. It is used in severe allergic asthma, chronic spontaneous urticaria and severe chronic rhinosinusitis with nasal polyps, where it reduces polyp size and improves symptoms and smell, though less dramatically than dupilumab in trials.

1 CHAPTER ONE At a glance

The drug on one page.

1.1 Summary

OMALIZUMAB

	SUMMARY
Class	Anti-IgE monoclonal antibody
How it works	Binds the Fc region of free IgE, reduces circulating IgE and downregulates high-affinity IgE receptors on mast cells and basophils
ENT uses	Severe chronic rhinosinusitis with nasal polyps; severe allergic asthma; chronic spontaneous urticaria
Dose	Subcutaneously every 2 or 4 weeks; for asthma and polyps the dose (75 to 600 mg) is set from a table by baseline total IgE and body weight; urticaria 300 mg every 4 weeks
Pregnancy	Use if clearly needed

2 CHAPTER TWO How it works

Mechanism and handling.

2.1 Mechanism

By removing free IgE it blunts both allergen-triggered and non-specific type 2 activation. Total IgE measured during treatment rises (complexes) and cannot be used to adjust the dose.

3 CHAPTER THREE Using it in ENT

Indications.

3.1 Polyposis

Eligible patients are as for other biologics (EPOS criteria): severe, type 2 polyposis not controlled by surgery and medical therapy. Response is assessed at 16 to 24 weeks.

4 CHAPTER FOUR Safety

What goes wrong.

4.1 Adverse effects

- **Anaphylaxis** in about 0.1% to 0.2%, sometimes delayed or after several doses: give the first doses where anaphylaxis can be treated, observe after injection, and consider an adrenaline auto-injector.
- Injection-site reactions, headache, arthralgia.
- Possible helminth infection risk; caution in parasitic areas.

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Mechanism (free IgE) and dosing by IgE and weight.
- Anaphylaxis risk and precautions.

5.2 Common pitfalls

COMMON PITFALLS

- Using total IgE during treatment to judge response.

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